

The Vision & Problem Statement

KOHLER-MITWPU AI Research Lab Challenge | Track 1: AI Bathroom Designer & Planner

The Industry Problem

- **Spatial Clearance Friction:** Constrained by strict building codes requiring 21–30" front spaces, plumbing wet/dry zoning, and unobstructed door swing corridors.
- **Aesthetic & Specification Fragmentation:** Severe difficulty matching brassware finishes, ceramics, and fixtures across disparate catalog lines without expensive consultation.
- **The AI Hallucination Barrier:** Standard LLMs lack innate spatial awareness, frequently hallucinating bounding boxes and overlapping physical fixtures in CAD generation.

The Proposed Solution

- **Neuro-Symbolic Design Architecture:** Fundamentally decouples creative design curation from underlying spatial geometry.
- **Deterministic 2D Space Planner:** Mathematically guarantees code-compliant, zero-clash fixture placement and dynamic door-swing clearance.
- **Generative Curation Consultant:** Powered by Gemini 3.6 Flash for professional material palettes, layered lighting strategies, and WaterSense eco-efficiency metrics.

Core Approach: Neuro-Symbolic Agentic Framework

Bridging Deterministic Spatial Physics with Generative Architectural Reasoning

Pillar 1: Symbolic Spatial & Clearance Engine

(The "Brainstem")

Typology Gating: Dynamically identifies room typology (e.g., distinguishing a sub-36 square foot Powder Room from an 80+ square foot Master Bath) to prune impossible fixtures prior to placement.

Wall-Slot Routing: Establishes isolated coordinate planes for wet zones (showers/tubs), dry zones (vanities), and privacy zones (toilets), guaranteeing zero spatial collision.

Entrance Envelopes: Computes dynamic dotted swing arcs across all four walls to maintain clear circulation paths.

Pillar 2: Priority-Based Knapsack Optimizer

(The "Accountant")

This financial logic layer iteratively prunes optional luxury fixtures.

Downgrades from a premium bathtub to an accessory, and finally to entry-tier alternatives.

Ensures the total price remains strictly under budget with zero financial deficit.

Pillar 3: Generative AI Architectural Consultant

(The "Architect")

Ingests the validated SKU bundle into Gemini 3.6 Flash to output bespoke design critiques.

Generates professional advice on sightlines and Kohler brassware harmony.

Curates sophisticated tile and stone pairings like Calacatta travertine with fluted walnut, and 2700K high-CRI lighting schemes.

System Architecture & Production Tech Stack

High-Throughput, Modular Pipeline Built for Millisecond Reactivity

VISUAL ARCHITECTURE FLOW



PRODUCTION TECH STACK BREAKDOWN

- **Frontend & State Management:** Built on Streamlit, utilizing custom responsive session state management and cached execution to ensure a seamless, low-latency user experience during iterative design changes.
- **Geometry & Drafting Engine:** Powered by a Matplotlib vector engine. It leverages specific matplotlib.patches (including Arc, Rectangle, and Circle) to programmatically render precise, scaled 2D floor plans and clearance zones.
- **LLM Engine:** Integrated with Google Gemini 3.6 Flash via the official google-genai SDK, ensuring rapid, high-context reasoning for design curation and material pairing.
- **Data Layer:** Operates on a deeply normalized JSON schema. This foundational layer houses exact physical dimensions, localized pricing in INR, aesthetic categorization tags, and critical WaterSense eco-ratings for every Kohler SKU.

Business ROI, Sustainability & Commercial Scalability

Aligning Intelligent Automation with Kohler's Design Excellence & ESG Goals

Accelerated Showroom Sales Cycle

This platform eliminates days of manual CAD drafting typically required during initial customer consultations.

Instantly delivers a downloadable Architectural Specification Sheet (.md) that is fully ready for contractors and seamlessly integrates with Kohler showroom inventory systems.

Commitment to Water Conservation

Addressing the 10% sustainability impact criteria, the system mandates WaterSense-compliant fixtures across all budget tiers.

Includes rimless dual-flush toilets operating at 3.0L/4.5L, faucets equipped with laminar aerators delivering 1.2 GPM (up to 32% water reduction), and Katalyst air-induction rainheads operating at 1.75 GPM to maximize pressure.

Omnichannel Commercial Deployment

The architecture is highly versatile and built for multi-platform integration across existing client touchpoints.

Designed to be readily embeddable as an interactive Kohler website widget, an in-store tablet kiosk, or an internal ERP quoting engine for enterprise developers.

Future Scalability & Lab Roadmap

Phase 2 will introduce automated 3D rendering via WebGL and Three.js for real-time visual exploration.

Phase 3 will integrate advanced computer vision, allowing users to upload a photograph or hand-drawn floor plan sketch to auto-populate room dimensions.